Reinforcement Learning - Homework 5

Exercise 2: Hands-on Gridworld

November 19, 2025

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1 Part 2.1(a): MazeGrid

Setup Values

- **Grid:** MazeGrid
- **Algorithm:** Q-learning
- **Episodes:** 100
- **Parameters:** $\epsilon = 0.3, \alpha = 0.5, \gamma = 0.9$, Noise $= 0.2$

Results

- **Q-Learning Average Returns:** 0.155
- **Q-Learning Start Value:** ≈ 0.33
- **Value Iteration Start Value:** 0.28 (converges to 1.00 at goal)

Answer

How are the learned values different from Value Iteration?

1. Value Iteration calculates the exact utility using the transition models that are known beforehand which results in a smooth gradient from the start state (0.28) to the goal (1.00). Since Q-learning uses exploration as well values are noisier and less consistent. The start state value (0.33) is close to the VI value, intermediate states fluctuate.

2. Q-learning relies on sampling stochastic transitions. With only 100 episodes and noise (noise = 0.2), the agent has not visited every state-action pair enough times to converge to the true mean.

3. The ϵ-greedy policy means the agent makes random actions 30% of the time, therefore preventing it from always following the optimal path during training, affecting the Q-value updates.

How can you make them closer to optimal values?

1. **Increase Episodes:** Running 500–1000 episodes would stablize more in the Q-value estimates by providing more samples.

2. **Decay Epsilon:** Gradually reducing ϵ over time would allow the agent to transition from exploration to exploitation, stabilizing more.

3. **Reduce Learning Rate (α):** A high learning rate (0.5) causes the values to oscillate with every new sample. Reducing α would stabilize more.

2 Part 2.1(b): BridgeGrid

Setup Values

- **Grid:** BridgeGrid
- **Noise:** 0.0 (Deterministic)
- **Episodes:** 100

Results

- **Q-Learning Start Value:** 0.90 (Drops to 0.00 immediately after)
- **Value Iteration Start Value:** 5.90 (Gradient increases to 10.00)
- **Observation:** The Q-learning agent fails to learn the path across the bridge, assigning 0.0 values to most intermediate states.

Answers

Comparison with Value Iteration

Value Iteration finds the optimal path across the bridge. Because of exploration the Q-learning agent, however, has a value of 0.90 at the start and 0.00 for most of the bridge, meaning it has not learned the safe path to the goal because of the cliffs.

Why does the agent usually not learn the optimal policy?

1. **The Exploration Issue:** The BridgeGrid represents a "narrow passage" problem. The optimal path is surrounded by terminal states with a reward of -100.
2. **Epsilon-Greedy Limitation:** With $\epsilon = 0.3$, the agent takes a random move 30% of the time. On a narrow bridge, a random move often results in falling off (reward -100).
3. **Negative Experience Dominance:** The agent encounters the negative reward (-100) far more frequently than the positive goal reward (10). These negative updates propagate backward, effectively teaching the agent that the bridge is dangerous. The agent learns to avoid the bridge entirely to minimize loss, rather than risking the crossing for the distant reward.

3 Part 2.1(c): CliffGrid

Setup Values

- **Grid:** CliffGrid
- **Episodes:** 300

Results

- **Learned Value for Start State ($V(s)$):** 1.85 (Positive)
- **Average Returns from Training:** -23.61 (Negative)

Answers

Why is there such a large difference?

This because of the difference between **Off-Policy** learning and **On-Policy** behavior.

1. **Learned Value (Off-Policy):** Q-learning approximates the value of the *optimal* policy. The value 1.85 represents the expected return if the agent was greedy. The agent knows the optimal path exists.

2. **Average Returns (On-Policy Behavior):** During training, the agent acts according to an ϵ-greedy policy. Even though it knows the optimal path is along the edge, it is forced to take random actions 30% of the time. When walking along the cliff edge, a random move "down" is catastrophic (-100 reward).

3. **Conclusion:** The learned value is what the agent may have gotten (Optimal Policy), while the average return is what the agent gets while exploring (Behavior Policy).

A Commands and Images

The following commands were used to generate the data for this report:

Listing 1: Commands

```
# Question 2.1(a): MazeGrid
python3 gridworld.py -g MazeGrid -a q -k 100 -q
python3 gridworld.py -g MazeGrid -a value -i 100

# Question 2.1(b): BridgeGrid
python3 gridworld.py -g BridgeGrid -a q -k 100 -n 0.0 -q
python3 gridworld.py -g BridgeGrid -a value -i 100 -n 0.0

# Question 2.1(c): CliffGrid
python3 gridworld.py -g CliffGrid -a q -k 300 -q
```

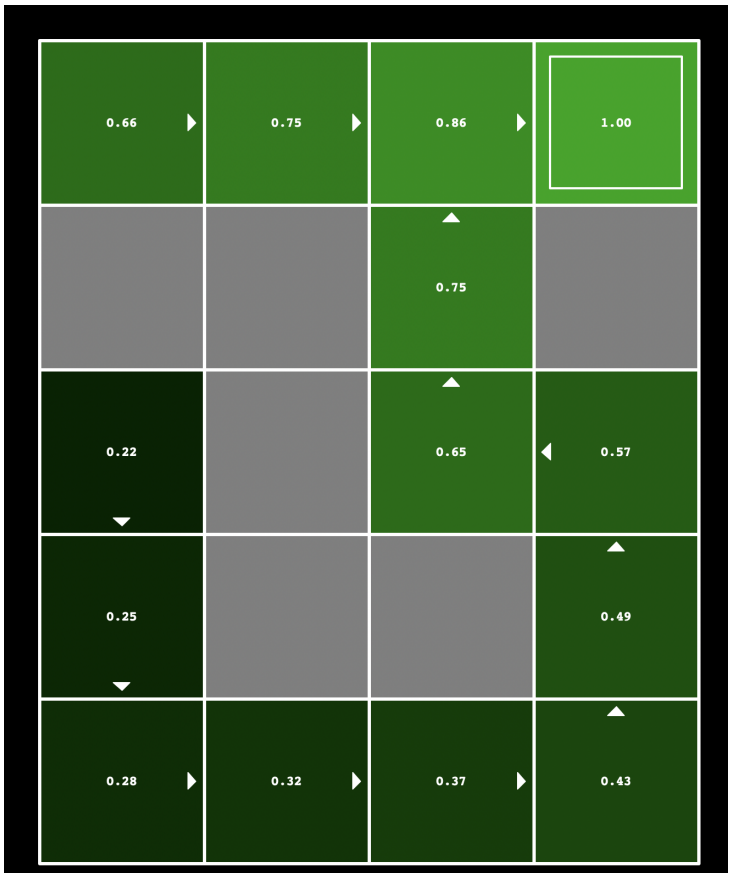

Figure 1: Maze Grid

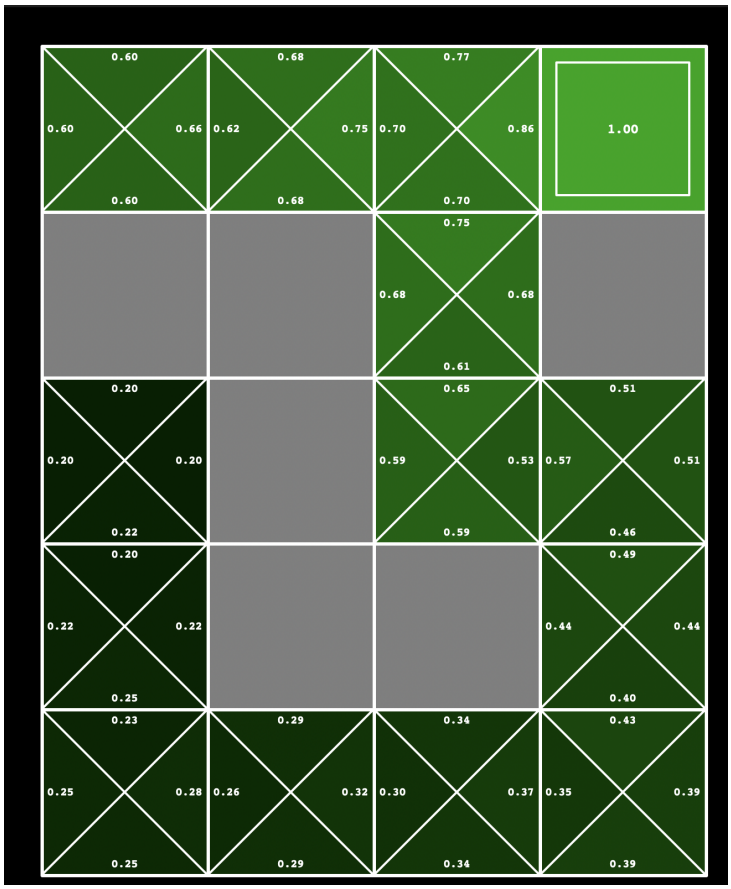

Figure 2: Maze Grid

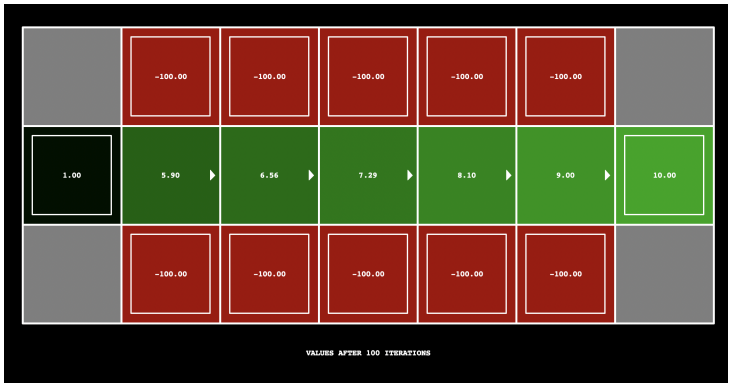

Figure 3: Bridge Grid

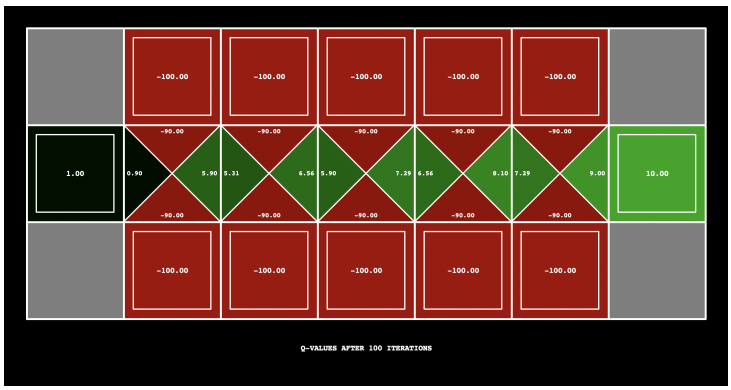

Figure 4: Bridge Grid